



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P3

Fill in the blanks to complete the notes on integration by substitution.

Part A

Indefinite integrals

We will illustrate the process for the integral $I = \int \frac{6x}{(2x-7)^{\frac{1}{2}}} dx$.

1. Define a new variable u . Let $u = 2x - 7$. Then $x =$.

2. Rewrite the integrand in terms of u .

$$I = \int \text{} dx = \int (3u^{\frac{1}{2}} + 21u^{-\frac{1}{2}}) dx$$

3. Differentiate $u = 2x - 7$ to get $\frac{du}{dx} =$, then make use of the chain rule to change the integral from an integral with respect to x into an integral with respect to u .

$$dx = \frac{dx}{du} du = \frac{1}{\frac{du}{dx}} du$$

$$\therefore I = \int \text{} du$$

4. Carry out the integration. This gives $I =$ $+ c$.

5. Finally, back-substitute for u to give the answer in terms of the original variable, x . This gives

$$I = \text{} + c.$$

Items:

Part B

Definite integrals

We will illustrate the process for the integral $J = \int_4^5 \frac{6x}{(2x - 7)^{\frac{1}{2}}} \, dx$. The first few steps are the same as for indefinite integrals.

- 1. Define a new variable u .
- 2. Rewrite the integrand in terms of u .
- 3. Use the chain rule to change the integral from an integral with respect to x into an integral with respect to u .

These steps give us

$$\therefore J = \int_{x=4}^{x=5} \boxed{} \, du$$

Next we consider the limits of the integral.

- 4. Use $u = 2x - 7$ to change the limits of the integral from values of x to the equivalent values of u .

$$J = \int_{u=\boxed{}}^{u=\boxed{}} \boxed{} \, du$$

- 5. Carry out the integration.

$$J = \left[\boxed{} \right]_{\boxed{}}^{\boxed{}}$$

Items:

1

2

3

$\frac{3}{2}u^{\frac{1}{2}} + \frac{21}{2}u^{-\frac{1}{2}}$

$u^{\frac{3}{2}} + 21u^{\frac{1}{2}}$



STEM SMART Single Maths 35 - Integration by Substitution >

Integration: General 4ii

Subject & topics: Maths

Stage & difficulty: A Level P3

Part A

Integrate $(4 - 3x)^7$

Find $\int (4 - 3x)^7 \, dx$.

The following symbols may be useful: c , x

Part B

Integrate $(4 - 3x)^{-1}$

Find $\int (4 - 3x)^{-1} \, dx$.

The following symbols may be useful: c , x

Used with permission from UCLES A-level Maths papers, 2003-2017.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution 5ii

Subject & topics: Maths **Stage & difficulty:** A Level P3

Use the substitution $u = 2x - 5$ to find the exact value of $\int_{\frac{5}{2}}^3 (4x - 8)(2x - 5)^7 dx$.

Used with permission from UCLES A-level Maths papers, 2003-2017.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution 5i

Subject & topics: Maths **Stage & difficulty:** A Level P3

Use the substitution $u = 2x + 1$ to evaluate $\int_0^{\frac{1}{2}} \frac{4x - 1}{(2x + 1)^5} dx$.

Used with permission from UCLES A-level Maths papers, 2003-2017.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution 2

Pre-Uni Maths for Sciences K3.2

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P3

Part A

Integrate $(bv + c)^2$

Find $\int (bv + c)^2 dv$, where b and c are constants.

The following symbols may be useful: b , c , k , v

Part B

Integrate $a(y - b)^3$

Find $\int_0^b a(y - b)^3 dy$, where a and b are constants.

The following symbols may be useful: a , b

Created for isaacphysics.org by Julia Riley.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution 1

Pre-Uni Maths for Sciences K3.1

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P3

Part A

Integrate $\sin(c\theta)$

Find $\int \sin(c\theta) \, d\theta$, where c is a constant.

The following symbols may be useful: C, c, cos(), k, sin(), tan(), theta

Part B

Integrate $e^{\alpha v}$

Find $\int e^{\alpha v} \, dv$, where α is a constant.

The following symbols may be useful: C, alpha, c, e, k, v

Created for isaacphysics.org by Julia Riley.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution 4ii

Subject & topics: Maths **Stage & difficulty:** A Level P3

Use the substitution $u = \sqrt{x+2}$ to find the exact value of $\int_{-1}^7 \frac{x^2}{\sqrt{x+2}} dx$.

Used with permission from UCLES A-level Maths papers, 2003-2017.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



STEM SMART Single Maths 35 - Integration by Substitution >

Integration by Substitution 4i

Subject & topics: Maths **Stage & difficulty:** A Level P3

Use the substitution $u = 1 + \sqrt{x}$ to find the exact value of $\int_4^9 \frac{1}{1 + \sqrt{x}} dx$ in the form $a + b \ln(c)$, where a , b , and c are positive constants to be found.

Used with permission from UCLES A-level Maths papers, 2003-2017.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution



Integration by Substitution 3

Pre-Uni Maths for Sciences K3.3

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P3

Part A

Integrate $\frac{3}{(z+1)^2}$

Find $\int_0^2 \frac{3}{(z+1)^2} \, dz$.

Part B

Integrate $\frac{e^{-\alpha x}}{(1+e^{-\alpha x})^4}$

Find $\int \frac{e^{-\alpha x}}{(1+e^{-\alpha x})^4} \, dx$, where α is a constant.

The following symbols may be useful: c, alpha, c, e, k, x

Created for isaacphysics.org by Julia Riley.

Question deck:

STEM SMART Single Maths 35 - Integration by Substitution